Sudarshan R. Hegde

Machine Learning Engineer | Bengaluru, India

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Professional Summary

Machine learning engineer (B.E. CSE, CGPA 9.1) with strong C++ and Python foundations and first-author published research in deep learning. Experienced in implementing, training and debugging models in PyTorch from reference architectures (GANs, UNet, CNN classifiers), validating results against baselines, and taking them through to deployed, full-stack systems. Currently exploring model deployment on resource-constrained hardware, and keen to work across the model, compiler and runtime stack to make models run fast and correctly on AI accelerators.

Technical Skills

Languages:	C++, Python, C, Java, JavaScript, TypeScript
ML & Deep Learning:	PyTorch, TensorFlow, Hugging Face, GANs, UNet, CNNs, Transformers (attention), active learning, Weights & Biases
Numerical & Vision:	NumPy, SciPy, OpenCV, tensor operations, model evaluation and numerical validation
Edge & Systems:	Embedded ML (ESP32-S3, Edge Impulse), Linux (Ubuntu), Docker, Git, Maven, Gradle
Serving & Web:	FastAPI, Node.js, Express.js, React, Tailwind CSS, REST APIs
Cloud & Tools:	AWS, Azure, Google Cloud, Jenkins, Kaggle GPUs, MySQL, LangChain (RAG), LaTeX

Work Experience

Product Development Intern, Continuum Associates LLC *Apr 2026 – Present*
Remote, India

- Build internal products and decision-support tools for clients in the electric power and energy sector, turning quantitative planning studies into reliable, performance-focused software features.
- Translate complex engineering rules (ISO, RTO and industry planning standards) into correct, testable implementations, working with cross-functional teams of planners, data and GIS specialists.
- Prototype, test and iterate on features with automated tests, prioritising correctness, performance and reliability.

AI with Cloud Computing Intern, SuprMentr Technologies Pvt Ltd *Jan 2026 – May 2026*
India

- Developed and deployed AI models in cloud-native environments, tuning for performance and cost-efficiency.
- Helped build automated data pipelines and integrate ML services into cloud platforms for high availability.

Creative Team Member, OSCode-SaIT Chapter *Sept 2025 – May 2026*
Bengaluru, India

- Contributed to open-source repositories through codebase maintenance, UI/UX design and technical documentation.

Projects

Geo-Agri Analyst *PyTorch, FastAPI, React, Hugging Face*

- Implemented and trained an RFB-ESRGAN super-resolution network for 4× upscaling of multi-spectral Sentinel-2 imagery, then a land-use classifier trained with Deep Bayesian active learning.
- Validated against BigEarthNet baselines: 5.4% higher Top-1 accuracy with roughly 85% fewer labels.
- Deployed the full pipeline as a web app with live satellite data, FastAPI inference backend and React front end.

Polygon Color Filling using UNet *PyTorch, Kaggle, W&B*

- Built a text-conditioned UNet that fills polygon shapes with specified colours, with custom data loading, training and evaluation tracked in Weights & Biases.

Smart Document Analyzer *Python, LangChain, RAG*

- Retrieval-augmented generation tool that answers questions over uploaded documents using LLMs.

Publications

Geo-Agri Analyst: An End-to-End Decision Support System for Agriculture via Super-Resolution and Active Learning, *International Journal of Scientific Development and Research (IJSDR)*, Vol. 11, Issue 2, Feb 2026. First author. Combined a perceptual super-resolution GAN with a hybrid active-learning query strategy. [Paper]

Education

B.E. in Computer Science and Engineering, Sambhram Institute of Technology (VTU) | CGPA 8.9/10 2022 – 2026
PUC (PCMCS), MES PU College, Sirsi | 88% 2020 – 2022

Certifications & Languages

- Deep Learning, IIT Ropar (NPTEL) | Robotic Process Automation (RPA) | Power BI and Data Visualization
- Languages: English, Kannada, Hindi